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Sec 3 Power

Assignment "3" Mechanics

1) $T_D = 15 + 2.2 \sin 2\theta - 1.85 \cos 2\theta$, $T_R = 15 \text{ kN.m}$

a) $\omega_{\text{mean}} = 2\pi \left(\frac{150}{60} \right) = 15.708 \text{ rad/s}$

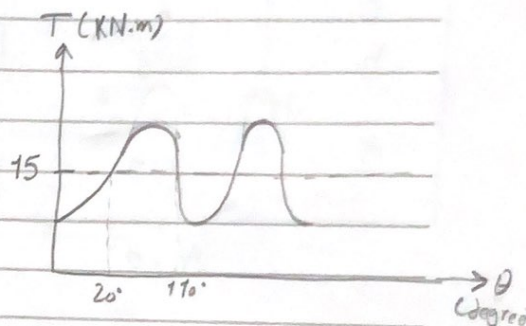
Power = $T_{\text{avg}} \times \omega_{\text{avg}} = 15 (15.708) = 235.62 \text{ Kwatt} = 315.97 \text{ hp}$

b) $\frac{\omega_{\text{max}} - \omega_{\text{min}}}{\omega_{\text{avg}}} = 0.01 = K_s$

$15 + 2.2 \sin 2\theta - 1.85 \cos 2\theta = 15$

$\frac{37}{44} \cos 2\theta = \sin 2\theta$, $\tan 2\theta = \frac{37}{44}$

$2\theta = 40^\circ$, $\theta = 20^\circ$, $2\theta = 40 + 180$, $\theta = 110^\circ$



$\Delta E_{\text{max}} = \int_{20^\circ}^{110^\circ} (2.2 \sin 2\theta - 1.85 \cos 2\theta) d\theta = 2.87 \text{ KJ}$

$\Delta E_{\text{max}} = I \omega_{\text{avg}}^2 K_s$, $2.87 \times 10^3 = (15.708)^2 (0.01) I$, $I = 1163.16 \text{ Kg.m}^2$

c) $T_D - T_R = I\alpha$, $\alpha = 0.8427 \text{ r/s}^2$

2) $K_s = 2 \times 0.025 = 0.05$, $\Delta E_{\text{max}} = \frac{3}{4} (T\theta)$, $\theta = \frac{2 \times 360}{180} T = 4\pi$

$P = T_{\text{avg}} \omega_{\text{avg}}$, $20 = 2\pi \left(\frac{200}{60} \right) T$, $T = 0.955 \text{ kN.m}$

$\Delta E_{\text{max}} = \frac{3}{4} (0.955) (4\pi) \times 10^3 = I \times K_s \left(\frac{2\pi \times 200}{60} \right)^2$, $I = 410.38 \text{ Kg.m}^2$

3) $T_s = 60$, $T_A = 100$, $N_s = 100 \text{ rpm}$, $C_{\text{Spider}} = 400 \text{ N.m}$

$N_A = 0$

SPider (Arm)	Sun (A)	Sun (S)
0	x	$-x \frac{T_A T_P}{T_P T_S}$
y	y	y
Σ	$x+y$	$y - x \frac{T_A}{T_S}$

$N_A = 0$

$N_s = 100$

$x+y=0$ (1) , $100 = y - x \frac{T_A}{T_S} \rightarrow 100 = y \left[1 + \frac{T_A}{T_S} \right]$

$y = 37.5 \text{ rpm}$, $\Sigma \text{Power} = 0$, $C_{\text{in}} \omega_{\text{in}} + C_{\text{out}} \omega_{\text{out}} = 0$

$C_{\text{in}} \omega_{\text{sun}} + C_{\text{Spider}} \omega_{\text{Spider}} = 0$, $C_{\text{sun}} \left(2\pi \left(\frac{100}{60} \right) \right) = -400 \left(2\pi \left(\frac{37.5}{60} \right) \right)$

$C_{\text{sun}} = -150.14 \text{ N.m}$, $\Sigma C = 0$

$C_{\text{sun}} + C_A + C_{\text{Spider}} = 0 \rightarrow C_A = -249.86 \text{ N.m}$

if $N_A = 20 \text{ rpm} \rightarrow x+y = 20$, $100 = y - x \frac{T_A}{T_S}$

$x = 20 - y$, $100 = y - (20 - y) \frac{T_A}{T_S} \rightarrow y = 50$

4) $N_E = 0$, $N_C = 0$, $T_C = 19$, $T_D = 18$, $T_E = 75$, $T_F = 74$, $T_A = T_B$

Arm (G)	E	A	F
O	x	$-x \frac{T_E T_B}{T_C T_A}$	$x \frac{T_E T_D}{T_C T_F}$
y	y	y	y
y	y+x	$y - x \frac{T_E T_B}{T_C T_A}$	$y + x \frac{T_E T_D}{T_C T_F}$

$$y + x = 0, N_E = 0, y \left[1 + \frac{T_E T_B}{T_C T_A} \right] = N_A, N_F = y \left[1 - \frac{T_E T_D}{T_C T_F} \right]$$

$$N_A = \frac{94}{19} y, N_F = \frac{28}{703} y, y = \frac{19}{94} N_A, N_F = \frac{28}{703} \cdot \frac{19}{94} N_A = \frac{-14}{1739} N_A \text{ (opposite in dir)}$$

$$\text{IF } T_E = 81, \frac{N_A}{N_F} = \frac{1 + \frac{T_E T_B}{T_C T_A}}{1 - \frac{T_E T_D}{T_C T_F}} = \frac{1850}{13}$$

5) $N_A = 1500 \text{ rpm}$

$$\frac{N_E}{N_A} = \frac{N_A}{N_P} \times \frac{N_P}{N_E}$$

$$\frac{N_E}{x} = - \frac{T_{P1} \cdot T_E}{T_{P2} \cdot T_A}$$

$$N_F = y = -1500 \text{ rpm}, N_F = -x \frac{48}{72}$$

$$\frac{N_E}{N_A} = \frac{N_D}{N_A} = \frac{N_D N_B}{N_C N_A} = \frac{T_A T_C}{T_B T_D}$$

$$N_E = -321.43 \text{ rpm}, -321.43 = x \cdot 1500 \rightarrow x = 1178.57 \text{ rpm}$$

$$N_G = y - x \left(\frac{48}{72} \right), N_G = -2285.71 \text{ rpm}$$

$$\frac{N_H}{N_{\text{shaft1}}} = \frac{-T_{\text{shaft1}}}{T_H}, N_{\text{shaft1}} = -979.59 \text{ rpm}, N_{\text{shaft2}} = 2742.85 \text{ rpm}$$